

```
import React, { useState, useEffect,
useCallback } from "react";
import { ChevronDown, ChevronUp, Check,
TrendingUp, TrendingDown, Circle,
CheckCircle2 } from "lucide-react";

const MODULES = [
  {
    id: "m1",
    ticker: "BASIC",
    title: "Market Foundations",
    blurb: "What a share actually is,
exchanges, and indices.",
    lessons: [
      { id: "l1", text: "A share is a unit
of ownership in a company – not a lottery
ticket." },
      { id: "l2", text: "Exchanges (NYSE,
NSE, LSE) are marketplaces; indices (S&P
500, Nifty 50) are baskets that track
them." },
      { id: "l3", text: "Order types:
market order (instant, any price) vs limit
order (your price, maybe never filled)." },
      { id: "l4", text: "Bid-ask spread is
the gap between what buyers offer and
sellers want – it's a hidden cost." },
    ],
    quiz: [
      {
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      q: "If you place a limit order
below the current market price to buy, what
happens?",
      options: ["It fills instantly at
market price", "It waits until the price
drops to your limit", "It's automatically
cancelled"],
      correct: 1,
    },
    {
      q: "An index like the S&P 500
represents:",
      options: ["A single company's
stock", "A basket of stocks tracked
together", "A type of bond"],
      correct: 1,
    },
    {
      q: "A wide bid-ask spread usually
signals:",
      options: ["High liquidity", "Low
liquidity or high uncertainty", "Guaranteed
profit"],
      correct: 1,
    },
  ],
},
{
  id: "m2",
  ticker: "FNDMT",
  title: "Fundamental Analysis",
  blurb: "Reading a company's financial
statements like a skeptic.",
  lessons: [
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    { id: "l1", text: "Income statement:  
is the company actually making money, or  
just growing revenue?" },  
    { id: "l2", text: "Balance sheet:  
what it owns vs what it owes – debt-to-  
equity matters more than people admit." },  
    { id: "l3", text: "P/E ratio tells  
you what you're paying per dollar of  
earnings – not whether it's 'cheap'." },  
    { id: "l4", text: "Free cash flow  
shows actual cash generated, harder to  
manipulate than reported earnings." },  
  ],  
  quiz: [  
    {  
      q: "A company with rising revenue  
but shrinking profit margins is most  
likely:",  
      options: ["Always a great buy",  
"Worth investigating – costs may be  
outpacing growth", "Guaranteed to fail"],  
      correct: 1,  
    },  
    {  
      q: "A high P/E ratio relative to  
peers usually means the market expects:",  
      options: ["Lower future growth",  
"Higher future growth", "Bankruptcy"],  
      correct: 1,  
    },  
    {  
      q: "Why do analysts often prefer  
free cash flow over net income?",  
      options: ["It's easier to
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manipulate", "It reflects real cash, less
affected by accounting choices", "It's
always higher"],
    correct: 1,
  },
],
},
{
  id: "m3",
  ticker: "CHART",
  title: "Technical Analysis",
  blurb: "Support, resistance, and
reading price action.",
  lessons: [
    { id: "l1", text: "Support and
resistance are price levels where
buying/selling pressure tends to cluster."
},
    { id: "l2", text: "Moving averages
smooth out noise to show trend direction,
not predict the future." },
    { id: "l3", text: "Volume confirms
moves – a breakout on low volume is often a
trap." },
    { id: "l4", text: "RSI and other
oscillators flag overbought/oversold
conditions, not exact turning points." },
  ],
  quiz: [
    {
      q: "A price breakout on unusually
low volume is generally a sign of:",
      options: ["Strong conviction",
"Possible false signal", "Guaranteed
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continuation"],
  correct: 1,
},
{
  q: "When a stock's price is consistently above its 50-day moving average, the trend is generally considered:",
  options: ["Bearish", "Bullish", "Irrelevant"],
  correct: 1,
},
{
  q: "An RSI reading above 70 typically suggests a stock may be:",
  options: ["Oversold", "Overbought", "Fairly valued"],
  correct: 1,
},
],
},
{
  id: "m4",
  ticker: "ALLOC",
  title: "Portfolio Building",
  blurb: "Diversification, allocation, and not betting it all on one name.",
  lessons: [
    { id: "l1", text: "Diversification reduces risk from any single company or sector blowing up." },
    { id: "l2", text: "Index funds give broad exposure with low fees – the boring choice that often wins." },
  ]
}
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{ id: "l3", text: "Dollar-cost
averaging removes the pressure of trying to
time entry perfectly." },
  { id: "l4", text: "Asset allocation
(stocks vs bonds vs cash) usually matters
more than individual stock picks." },
],
quiz: [
  {
    q: "Why might a low-cost index fund
outperform many actively picked portfolios
over time?",
    options: ["It avoids fees and
stock-picking errors that compound against
you", "It only buys stocks guaranteed to
rise", "It has insider access"],
    correct: 0,
  },
  {
    q: "Dollar-cost averaging works
by:",
    options: ["Timing the exact
bottom", "Investing fixed amounts at
regular intervals regardless of price",
"Only buying during crashes"],
    correct: 1,
  },
  {
    q: "Which decision tends to drive
long-run portfolio outcomes the most?",
    options: ["Picking the single best
stock", "Overall asset allocation",
"Trading frequency"],
    correct: 1,
  }
]
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    },
  ],
},
{
  id: "m5",
  ticker: "PSYCH",
  title: "Market Psychology",
  blurb: "The biases that wreck good
plans.",
  lessons: [
    { id: "l1", text: "Loss aversion:
losses hurt roughly twice as much as
equivalent gains feel good." },
    { id: "l2", text: "FOMO buying near
the top is one of the most common ways
money is lost." },
    { id: "l3", text: "Herd mentality
feels safe in the moment and is often wrong
at the extremes." },
    { id: "l4", text: "Confirmation bias
makes you seek news that supports a
position you already hold." },
  ],
  quiz: [
    {
      q: "Selling a winning stock too
early and holding a losing one too long is
a classic symptom of:",
      options: ["Good discipline", "Loss
aversion bias", "Diversification"],
      correct: 1,
    },
    {
      q: "Buying a stock mainly because

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'everyone is talking about it' is an
example of:",
    options: ["Fundamental analysis",
"Herd mentality / FOMO", "Risk
management"],
    correct: 1,
  },
  {
    q: "Only reading news that confirms
your existing investment thesis is
called:",
    options: ["Confirmation bias",
"Diversification", "Technical analysis"],
    correct: 0,
  },
],
},
{
  id: "m6",
  ticker: "RISK",
  title: "Risk & Position Sizing",
  blurb: "How much to put on any single
bet.",
  lessons: [
    { id: "l1", text: "Position sizing
controls how much any single loss can hurt
your overall portfolio." },
    { id: "l2", text: "A stop-loss is a
pre-decided exit – deciding it in advance
removes emotion from the moment." },
    { id: "l3", text: "Risking a small,
fixed percentage per trade keeps one bad
call from being fatal." },
    { id: "l4", text: "Correlated
```



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positions (e.g. five tech stocks) behave
like one big position in a downturn." },
],
quiz: [
  {
    q: "The main purpose of position
sizing is to:",
    options: ["Maximize the size of
every trade", "Limit the damage any single
position can do", "Guarantee profits"],
    correct: 1,
  },
  {
    q: "Setting a stop-loss before
entering a trade primarily helps you:",
    options: ["Avoid emotional
decisions in the moment", "Increase
position size", "Avoid taxes"],
    correct: 0,
  },
  {
    q: "Owning five highly correlated
tech stocks instead of one is:",
    options: ["True diversification",
"Effectively one concentrated bet", "Risk-
free"],
    correct: 1,
  },
],
},
{
  id: "m7",
  ticker: "DIVTX",
  title: "Dividends & Taxes",
```

```
    blurb: "Getting paid to hold, and what
the taxman wants.",
    lessons: [
      { id: "l1", text: "Dividends are a
portion of profit paid to shareholders –
not all companies pay them." },
      { id: "l2", text: "Dividend yield is
annual dividend ÷ share price; a yield that
looks too high can signal trouble." },
      { id: "l3", text: "Long-term capital
gains (held over a year in most
jurisdictions) are usually taxed lower than
short-term." },
      { id: "l4", text: "Reinvesting
dividends compounds returns significantly
over decades." },
    ],
    quiz: [
      {
        q: "An unusually high dividend
yield compared to peers often signals:",
        options: ["A clearly safe bet", "A
falling stock price or financial trouble",
"Nothing meaningful"],
        correct: 1,
      },
      {
        q: "In most tax systems, holding a
stock longer than a year before selling
tends to:",
        options: ["Increase your tax rate
on gains", "Lower your tax rate on gains",
"Have no tax effect"],
        correct: 1,
```

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    },
    {
      q: "Reinvesting dividends instead
of cashing them out mainly:",
      options: ["Has no long-term
effect", "Compounds your returns over
time", "Increases your tax bill
immediately"],
      correct: 1,
    },
  ],
},
{
  id: "m8",
  ticker: "IPOLST",
  title: "IPOs & New Listings",
  blurb: "Why the hype around new stocks
can be a trap.",
  lessons: [
    { id: "l1", text: "An IPO is when a
private company sells shares to the public
for the first time." },
    { id: "l2", text: "Early IPO pops are
often driven by limited float and hype, not
fundamentals." },
    { id: "l3", text: "Lock-up periods
prevent insiders from selling immediately –
expiry can pressure the price." },
    { id: "l4", text: "Newly listed
companies often lack a long earnings track
record to analyze." },
  ],
  quiz: [
    {
```

```
      q: "A large first-day price 'pop'
in an IPO is mainly explained by:",
      options: ["Proven long-term
fundamentals", "Limited share supply and
hype-driven demand", "Government
guarantee"],
      correct: 1,
    },
    {
      q: "A lock-up period expiring often
causes the stock price to:",
      options: ["Rise due to new buyers",
"Face downward pressure as insiders can now
sell", "Stay completely flat"],
      correct: 1,
    },
  ],
},
{
  id: "m9",
  ticker: "GLOBL",
  title: "Macro & Global Markets",
  blurb: "Interest rates, inflation, and
why the whole market moves together
sometimes.",
  lessons: [
    { id: "l1", text: "Rising interest
rates generally make borrowing costlier and
can pressure stock valuations." },
    { id: "l2", text: "Inflation erodes
purchasing power and affects which sectors
perform well." },
    { id: "l3", text: "Currency moves
affect multinational companies' reported
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earnings." },
    { id: "l4", text: "Markets often move
on expectations of news, not just the news
itself." },
  ],
  quiz: [
    {
      q: "When central banks raise
interest rates, stock valuations often:",
      options: ["Rise, since borrowing is
cheaper", "Face downward pressure", "Are
completely unaffected"],
      correct: 1,
    },
    {
      q: "A US company earning revenue
overseas is affected by currency moves
because:",
      options: ["Foreign earnings must be
converted back, changing reported results",
"Currency has no link to earnings", "Only
bonds are affected by currency"],
      correct: 0,
    },
    {
      q: "Markets reacting before an
expected event (like a rate decision)
actually happens illustrates that:",
      options: ["Markets only react after
news", "Prices often move on anticipation,
not just outcomes", "Macro events don't
affect stocks"],
      correct: 1,
    },
  ],
}
```

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    ],
  },
];

const STORAGE_KEY = "marketschool-progress-
v2";

function loadProgress(setProgress,
setLoading) {
  window.storage
    .get(STORAGE_KEY, false)
    .then((res) => {
      if (res && res.value)
setProgress(JSON.parse(res.value));
      setLoading(false);
    })
    .catch(() => setLoading(false));
}

function saveProgress(progress) {
  window.storage.set(STORAGE_KEY,
JSON.stringify(progress), false).catch(()
=> {});
}

function moduleStats(module, state) {
  const total = module.lessons.length +
module.quiz.length;
  const lessonsDone =
module.lessons.filter((l) =>
state?.lessons?.[l.id]).length;
  const quizDone = module.quiz.filter((_,
i) => state?.quiz?.[i] === true).length;
  const done = lessonsDone + quizDone;

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    return { total, done, pct:
Math.round((done / total) * 100) };
}

function TickerTape({ modules, progress })
{
    const items = modules.map((m) => ({
ticker: m.ticker, pct: moduleStats(m,
progress[m.id]).pct }));
    const loop = [...items, ...items];
    return (
        <div className="overflow-hidden border-
y border-[#1E2430] bg-[#0B0E14] py-2.5">
            <div className="flex w-max animate-
[scroll_34s_linear_infinite] gap-8 px-4">
                {loop.map((it, i) => (
                    <div key={i} className="flex
items-center gap-1.5 font-mono text-[13px]
whitespace-nowrap">
                        <span className="text-
[#8B92A5]">{it.ticker}</span>
                        <span className={it.pct > 0 ?
"text-[#00C896]" : "text-[#8B92A5]}">
                            {it.pct > 0 ? <TrendingUp
size={12} className="inline -mt-0.5" /> :
<TrendingDown size={12} className="inline -
mt-0.5" />}
                        </span>
                        <span className={it.pct === 100
? "text-[#00C896]" : "text-[#E8EAED]}">
{it.pct}%</span>
                    </div>
                ))}
            </div>
        </div>
    )
}

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    <style>{@keyframes scroll { from {
transform: translateX(0); } to { transform:
translateX(-50%); } }`</style>
  </div>
  );
}

function ProgressChart({ modules, progress
}) {
  const points = modules.reduce(
    (acc, m) => {
      const { pct } = moduleStats(m,
progress[m.id]);
      const last = acc[acc.length - 1];
      acc.push(last + pct / 100);
      return acc;
    },
    [0]
  );
  const max = modules.length;
  const w = 600,
    h = 120,
    pad = 12;
  const coords = points.map((p, i) => {
    const x = pad + (i / (points.length -
1)) * (w - pad * 2);
    const y = h - pad - (p / max) * (h -
pad * 2);
    return [x, y];
  });
  const path = coords.map(([x, y], i) => (i
=== 0 ? `M${x},${y}` :
`L${x},${y}`)).join(" ");
  const areaPath = `${path}

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L${coords[coords.length - 1][0]},${h - pad}
L${coords[0][0]},${h - pad} Z`;
    const overallPct =
Math.round((points[points.length - 1] /
max) * 100);

    return (
      <div className="rounded-lg border
border-[#1E2430] bg-[#131720] p-5">
        <div className="mb-1 flex items-
baseline justify-between">
          <span className="font-mono text-xs
uppercase tracking-wider text-
[#8B92A5]">Your Progress · LRN</span>
          <span className="font-mono text-2xl
font-semibold text-[#00C896]">{overallPct}%
</span>
        </div>
        <svg viewBox={`0 0 ${w} ${h}`}
className="w-full"
preserveAspectRatio="none">
          <defs>
            <linearGradient id="fillGrad"
x1="0" y1="0" x2="0" y2="1">
              <stop offset="0%"
stopColor="#00C896" stopOpacity="0.25" />
              <stop offset="100%"
stopColor="#00C896" stopOpacity="0" />
            </linearGradient>
          </defs>
          <path d={areaPath}
fill="url(#fillGrad)" />
          <path d={path} fill="none"
stroke="#00C896" strokeWidth="2" />

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        {coords.map(([x, y], i) => (
          <circle key={i} cx={x} cy={y}
r="3" fill="#0B0E14" stroke="#00C896"
strokeWidth="2" />
        )))
      </svg>
    </div>
  );
}

function QuizQuestion({ q, index,
savedCorrect, onAnswer }) {
  const [selected, setSelected] =
useState(null);
  const showResult = savedCorrect !==
undefined && savedCorrect !== null ?
savedCorrect : selected !== null ? selected
=== q.correct : null;

  return (
    <div className="rounded-md bg-[#0B0E14]
border border-[#1E2430] p-4">
      <div className="mb-2.5 text-sm font-
medium text-[#E8EAED]">
        <span className="text-[#5A6177] mr-
1.5 font-mono text-xs">Q{index + 1}</span>
        {q.q}
      </div>
      <div className="space-y-1.5">
        {q.options.map((opt, i) => {
          const isSelected = selected ===
i;
          const isCorrect = i ===
q.correct;

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        let style = "border-[#1E2430]
text-[#E8EAED] hover:border-[#3A4254]";
        const answered = selected !==
null || savedCorrect !== undefined;
        if (answered) {
            if (isCorrect) style = "border-
[#00C896]/50 bg-[#00C896]/10 text-
[#00C896]";
            else if (isSelected) style =
"border-[#FF5C5C]/50 bg-[#FF5C5C]/10 text-
[#FF5C5C]";
            else style = "border-[#1E2430]
text-[#5A6177]";
        }
        return (
            <button
                key={i}
                onClick={() => {
                    setSelected(i);
                    onAnswer(i === q.correct);
                }}
                className={`flex w-full
items-center justify-between rounded-md
border px-3 py-2 text-left text-[13px]
transition-colors ${style}`}
            >
                <span>{opt}</span>
                {answered && isCorrect &&
<Check size={14} />}
            </button>
        );
    }
}
</div>
</div>

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    );
}

function ModuleCard({ module, state,
onToggleLesson, onAnswerQuiz, expanded,
onExpand }) {
    const { total, done, pct } =
moduleStats(module, state);
    const complete = pct === 100;

    return (
        <div className={`rounded-lg border bg-
[#131720] transition-colors ${complete ?
"border-[#00C896]/40" : "border-
[#1E2430]"}`}>
            <button onClick={onExpand}
className="flex w-full items-center
justify-between gap-3 px-5 py-4 text-left">
                <div className="flex items-center
gap-3 min-w-0">
                    <span className={`font-mono text-
xs font-semibold rounded px-1.5 py-0.5
${complete ? "bg-[#00C896]/15 text-
[#00C896]" : "bg-[#1E2430] text-
[#8B92A5]"}`}>
                        {module.ticker}
                    </span>
                    <div className="min-w-0">
                        <div className="truncate text-
[15px] font-medium text-[#E8EAED]">
{module.title}</div>
                        <div className="truncate text-
[13px] text-[#8B92A5]">{module.blurb}</div>
                    </div>
                </div>
            </button>
        </div>
    );
}

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    </div>
    <div className="flex items-center
gap-3 shrink-0">
      <span className={`font-mono text-
sm ${complete ? "text-[#00C896]" : "text-
[#8B92A5]"}`}>
        {done}/{total}
      </span>
      {expanded ? <ChevronUp size={18}
className="text-[#8B92A5]" /> :
<ChevronDown size={18} className="text-
[#8B92A5]" />}
    </div>
  </button>

  {expanded && (
    <div className="border-t border-
[#1E2430] px-5 py-4 space-y-4">
      <div className="space-y-2">
        {module.lessons.map((l) => {
          const checked =
!!state?.lessons?.[l.id];
          return (
            <button
              key={l.id}
              onClick={() =>
onToggleLesson(l.id)}
              className="flex w-full
items-start gap-2.5 rounded-md px-2 py-1.5
text-left hover:bg-[#1A2030] transition-
colors"
            >
              {checked ? (
                <CheckCircle2 size={17}

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        className="mt-0.5 shrink-0 text-[#00C896]"
      />

      ) : (
        <Circle size={17}
        className="mt-0.5 shrink-0 text-[#3A4254]"
      />

      )}
      <span className={`text-sm
      ${checked ? "text-[#8B92A5] line-through
      decoration-[#3A4254]" : "text-
      [#E8EAE8]"}`}>
        {l.text}
      </span>
    </button>
  );
  })}
</div>

  <div className="space-y-3">
    {module.quiz.map((q, i) => (
      <QuizQuestion
        key={i}
        q={q}
        index={i}
        savedCorrect={state?.quiz?.
[i]}
        onAnswer={({correct) =>
onAnswerQuiz(i, correct)}
      />
    )})}
  </div>
</div>
)}
</div>

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    );
  }

export default function MarketSchool() {
  const [progress, setProgress] =
  useState({});
  const [loading, setLoading] =
  useState(true);
  const [expandedId, setExpandedId] =
  useState(MODULES[0].id);

  useEffect(() => {
    loadProgress(setProgress, setLoading);
  }, []);

  const update = useCallback((updater) => {
    setProgress((prev) => {
      const next = updater(prev);
      saveProgress(next);
      return next;
    });
  }, []);

  const toggleLesson = (moduleId, lessonId)
=> {
    update((prev) => {
      const mod = prev[moduleId] || {
lessons: {}, quiz: {} };
      return { ...prev, [moduleId]: {
...mod, lessons: { ...mod.lessons,
[lessonId]: !mod.lessons[lessonId] } } };
    });
  };

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```

    const answerQuiz = (moduleId, quizIndex,
correct) => {
      update((prev) => {
        const mod = prev[moduleId] || {
lessons: {}, quiz: {} };
        return { ...prev, [moduleId]: {
...mod, quiz: { ...mod.quiz, [quizIndex]:
correct } } };
      });
    };

    const totalLessons = MODULES.reduce((s,
m) => s + m.lessons.length, 0);
    const totalQuiz = MODULES.reduce((s, m)
=> s + m.quiz.length, 0);

    if (loading) {
      return (
        <div className="flex min-h-screen
items-center justify-center bg-[#0B0E14]">
          <span className="font-mono text-sm
text-[#8B92A5] animate-pulse">loading
market data...</span>
        </div>
      );
    }

    return (
      <div className="min-h-screen bg-
[#0B0E14] text-[#E8EAED]">
        <header className="px-5 pt-7 pb-5">
          <div className="mx-auto max-w-2xl">
            <div className="flex items-center
gap-2 mb-1">

```



```

        <span className="h-2 w-2
rounded-full bg-[#00C896]" />
        <span className="font-mono
text-[11px] uppercase tracking-[0.2em]
text-[#8B92A5]">Live · Self-paced</span>
      </div>
      <h1 className="text-[28px] font-
semibold tracking-tight">MarketSchool</h1>
      <p className="mt-1 text-sm text-
[#8B92A5]">
        {MODULES.length} modules ·
        {totalLessons} lessons · {totalQuiz} quiz
        questions. Your progress moves like a price
        chart.
      </p>
    </div>
  </header>

  <TickerTape modules={MODULES}
progress={progress} />

  <main className="mx-auto max-w-2xl
px-5 py-6 space-y-5">
    <ProgressChart modules={MODULES}
progress={progress} />

    <div className="space-y-3">
      {MODULES.map((m) => (
        <ModuleCard
          key={m.id}
          module={m}
          state={progress[m.id]}
          expanded={expandedId ===
m.id}

```

```

        onExpand={() =>
setExpandedId(expandedId === m.id ? null :
m.id)}
        onToggleLesson={() =>
toggleLesson(m.id, lessonId)}
        onAnswerQuiz={() =>
=> answerQuiz(m.id, qIdx, correct)}
        />
    ))}
</div>

    <p className="pt-2 pb-8 text-center
text-[11px] font-mono text-[#5A6177]">
        progress saved automatically ·
this is education, not financial advice
    </p>
</main>
</div>
);
}

```